

David Ettel

EDUCATION

Harvard University

Bachelor of Arts in Mathematics, Minor in Computer Science, GPA: 3.91

Sept. 2022 – May 2026

Cambridge, MA

Relevant Coursework

Abstract Algebra (Math55A), Topology & Real and Complex Analysis (Math55B), Algebraic Topology (Graduate), Advanced Complex Analysis (Graduate), Differential Topology, Algebraic Geometry, Algebraic Number Theory, Measure Theory, Algebra II, Computational Properties in Interpretable Machine Learning (Graduate), Introduction to Machine Learning (MIT), Introduction to Probability, Algorithms and Data Structures, Systems Programming and Machine Organization, Geometric Methods in Machine Learning (Graduate), Physical Mathematics (Graduate), Quantum Learning Theory (Graduate), Game Theory and Applications

Sir Karl Popper School

High School

Sept. 2017 – June 2021

Vienna, Austria

EXPERIENCE

Mathematical Researcher, Graph Theory

Philip Matchett Wood Group, Harvard University

June 2025 – August 2025

Cambridge, MA

- Designing and leading an original research project on advanced topics in graph theory as principal investigator.
- Conducting research on a conjecture proposed in 2025, addressing one of the newest open problems in contemporary graph theory.
- Engaged in 60+ hours per week of mathematical problem-solving, theorem development, and literature synthesis.
- Preparing findings for later submission to a peer-reviewed mathematics journal.

Research Assistant, Almost Equivariant Machine Learning

Melanie Weber Group, Harvard University

Oct. 2024 – Present

Cambridge, MA

- Principal investigator on a 6k Harvard KURE grant.
- Generalize results from equivariant machine learning to the almost equivariant case.
- Develop experiments to investigate the scalability of almost equivariant machine learning.

Research Assistant, Topological Deep Learning

Vikas Garg Group, AALTO University

June 2024 – Aug. 2024

Espoo, Finland

- Developed an original research project on topological deep learning, focusing on neural approximators for persistent homotopy. Conducted an extensive literature review to formulate a research question in previously unmapped territory.
- Built the theoretical foundation for the expressiveness of the approximators and designed the corresponding network architecture.
- Selected from 4,300 competitive applicants, passing a highly selective interview process that admitted only 60 candidates.

Teaching Assistant, Pure & Applied Mathematics

Department of Mathematics & SEAS, Harvard University

Jan. 2024 – Present

Cambridge, MA

- Assist with teaching Physical Mathematics (Graduate), Differential Topology, Real Analysis, Algebra I, Classical Geometry.
- Grade problem sets for up to 80 students weekly, improving the ability to evaluate logical reasoning for correctness.
- Guide students through challenging concepts during office hours, enhancing skills in communicating abstract ideas.
- Teach additional topics and review course material in section.

Alternative Military Service, Kindergarten Assistant

City of Vienna

Oct. 2021 – June 2022

Cambridge, MA

- Instructed and supervised 1 to 10 year olds. Twenty at a time for 40 hours per week. Over 200 children overall.
- Instructing included speaking, walking, reading, counting, math, chess, and motor skill development.
- Gained extensive experience in group leading and event planning. Developed strong didactic skills.
- Other responsibilities included food preparation and cleaning.

PUBLICATIONS

Effects of Network Pruning on LLM Interpretability

Spring 2025

Published in The Harvard Undergraduate Research Journal (THURJ)

Cambridge, MA

- Co-authored published research analyzing the impact of network pruning on SHAP and IG explanation faithfulness in DistilBERT and RoBERTa.
- Found that explanation reliability is more sensitive to architecture and pruning randomness than to magnitude-based sparsity.
- Investigated geometric underpinnings of interpretability via Hessian-based curvature analysis.
- Read online: <https://doi.org/10.62571/gujf5t1klb25jbcdbde>
- GitHub Code Repository: <https://github.com/camilobrownpinilla/Explain-This-Pruner>

HARDMath2: A Benchmark for Applied Mathematics

Submitted May 2025

Accepted at Datasets & Benchmarks Track, NeurIPS 2025

San Diego, CA & Mexico City, MX

- Co-wrote background & experimental-setup sections and helped shape coverage of boundary-layer analysis, WKB methods, nonlinear-PDE asymptotics, and oscillatory-integral approximations.
- Authored 10 graduate-level benchmark problems—each requiring multi-step asymptotic analysis—and peer-verified 30+ solutions.
- Sole student member on the rebuttal team, coordinating point-by-point responses to reviewer feedback under 6-day deadlines.
- Arxiv link: <https://arxiv.org/abs/2505.11774>

EXTRACURRICULAR ACTIVITIES

Technical Fellow

Sept. 2024 – Present

Harvard AI Safety Club

Cambridge, MA

- Discuss technical AI safety papers in weekly meetings.
- Attend monthly workshops with industry experts from Redmond Research, OpenAI, Anthropic, and other companies.

Problem Writer and Tester

Sept. 2023 – May 2024

Harvard-MIT Math Tournament

Cambridge, MA

- Invented and developed problems for the biannual Harvard-MIT Math Tournament, contributing to the participation of 1,000+ high school students from around the world.
- Evaluated and solved test problems for the Harvard-MIT Math Tournament, ensuring accuracy and appropriate difficulty levels for 76 questions posed in each competition.
- Graded 700 exams on competition day, ensuring timely and accurate assessment of participant performance.

AWARDS

Mathematics Olympiads

National & International

- Achieved Honorable Mention and second-best Austrian result at Cyber Mathematical Competition 2020. (compensatory competition for deferred IMO 2020, each country sent their best eight students.)
- Achieved Honorable Mention and best Austrian result at Middle European Mathematics Olympiad 2020. (competition to which European countries send their 6 best students not in IMO team.)
- Achieved Top Ten Placements in national finals 2020 and 2021.

”Pre-scientific paper”, part of Austrian Matura (VWA)

”Comparison of the geometries resulting from a modification of the Euclidean Parallel Postulate”

- 1st place at the “Hans-Riegel-Fachpreis” for mathematical pre-scientific papers, awarded by University of Vienna.
- Best mathematical pre-scientific paper in Austria award from Austrian Mathematical Society (ÖMG) in 2021

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS, Bash/Shell

Developer Tools: Git, Docker, VS Code, PyCharm, Jupyter, Colab, Weights & Biases, $\text{\LaTeX}$

Libraries: PyTorch, PyTorch Geometric, NumPy, Matplotlib, SymPy, NetworkX

LANGUAGES

German: Native Speaker.

English: C2.

Italian: B1.

French: A1.